

User behaviour analytics

```
import hashlib
```

```
import json
```

```
import os
```

```
import time
```

```
from datetime import datetime
```

```
PROTECTED_FILES = [
```

```
    "protected_file.txt",
```

```
    "config.json"
```

```
]
```

```
BASELINE_FILE = "baseline_hashes.json"
```

```
LOG_FILE = "tamper_log.txt"
```

```
# Track files already reported as deleted
```

```
deleted_files = set()
```

```
def get_file_hash(filepath):
```

```
    sha256 = hashlib.sha256()
```

```
    with open(filepath, "rb") as f:
```

```
        while chunk := f.read(4096):
```

```
            sha256.update(chunk)
```

```
return sha256.hexdigest()
```

```
def create_baseline():
```

```
    hashes = {}
```

```
    for file in PROTECTED_FILES:
```

```
        if os.path.exists(file):
```

```
            hashes[file] = get_file_hash(file)
```

```
    with open(BASELINE_FILE, "w") as f:
```

```
        json.dump(hashes, f, indent=4)
```

```
    print("Baseline Created")
```

```
def load_baseline():
```

```
    with open(BASELINE_FILE, "r") as f:
```

```
        return json.load(f)
```

```
def save_baseline(baseline):
```

```
    with open(BASELINE_FILE, "w") as f:
```

```
        json.dump(baseline, f, indent=4)
```

```
def log_event(message):

    timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

    log = f"[{timestamp}] {message}"

    print(log)

    with open(LOG_FILE, "a") as f:
        f.write(log + "\n")

# Create baseline if it doesn't exist
if not os.path.exists(BASELINE_FILE):
    create_baseline()

baseline = load_baseline()

print("Tamper Protection Started...")
print("-----")

while True:

    for file in PROTECTED_FILES:

        # File deleted
        if not os.path.exists(file):
```

```
if file not in deleted_files:
```

```
    log_event(f"TAMPER ALERT: {file} deleted!")
```

```
    deleted_files.add(file)
```

```
    continue
```

```
# File restored
```

```
if file in deleted_files:
```

```
    deleted_files.remove(file)
```

```
    current_hash = get_file_hash(file)
```

```
    baseline[file] = current_hash
```

```
    save_baseline(baseline)
```

```
    log_event(f"INFO: {file} restored and monitoring resumed.")
```

```
    continue
```

```
current_hash = get_file_hash(file)
```

```
# New file added to baseline
```

```
if file not in baseline:
```

```
    baseline[file] = current_hash
```

```
    save_baseline(baseline)
```

```
    continue
```

```
# File modified
```

```
if current_hash != baseline[file]:
```

```
    log_event(f"TAMPER ALERT: {file} modified!")
```

```
    baseline[file] = current_hash
```

```
    save_baseline(baseline)
```

```
time.sleep(5)
```